

Municipal Solid Waste Management Using Machine Learning: A Case Study in Sheger City, Koye Sub-city, Ethiopia

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Article History:

Received: 08.03.2025 Revised: 28.03.2025 Accepted: 01.04.2025 Published: 11.04.2025

Abstract

Municipal Solid Waste Management is an increasingly critical challenge in urban areas, intensified by rapid urbanization, population growth, and evolving consumption patterns. This study investigates the application of machine learning techniques to predict municipal solid waste generation in Sheger City, Koye Sub-city, Ethiopia, using data from 2009 to 2023. Three machine learning models, ARIMA, RF, and LSTM, were employed to forecast waste generation trends for the period 2024–2028, considering various socio-economic and demographic factors. Among the models, LSTM demonstrated the highest accuracy, with MSE of 1.62×10^8 tonnes, MAE of 9,500 tonnes, and R^2 of 0.93. These results outperformed ARIMA (MSE = 3.84×10^8 tonnes², MAE = 15,200 tonnes, R^2 = 0.85) and RF (MSE = 2.91×10^8 tonnes², MAE = 12,800 tonnes, R^2 = 0.89). The forecasts predict an 8.5% increase in total waste generation, from 3,852,150 tonnes in 2023 to 4,177,500 tonnes by 2028. Notable growth is expected in high-volume waste streams, including food waste (13.5% increase) and plastic waste (8.9% increase). These findings highlight the urgent need for enhanced waste management strategies, including expanded recycling programs and policy interventions. This study provides a robust framework for leveraging machine learning models to guide waste management decisions, contributing to more sustainable urban waste management practices in rapidly growing cities.

Keywords: Solid Waste Management, Machine Learning, Random Forest, Waste Generation, Waste Prediction.

Suggested citation: Gudeta, T.H., Mamo, G.T., & Neguse, Y.M. (2025). Municipal Solid Waste Management Using Machine Learning: A Case Study in Sheger City, Koye Sub-city, Ethiopia. *European Journal of Theoretical and Applied Sciences*, 3(2), 511-525. [https://doi.org/10.59324/ejtas.2025.3\(2\).42](https://doi.org/10.59324/ejtas.2025.3(2).42)

Introduction

Municipal Solid Waste Management (MSWM) is a critical concern in urban areas, particularly as cities worldwide experience rapid population

growth and urbanization. The World Bank estimates that by 2025, urban areas will generate approximately 2.2 billion tons of waste annually, with developing countries facing the most significant challenges in managing this waste



effectively (Alshaikh & Abdelfatah, 2024). Inefficient waste management systems can lead to adverse environmental and public health outcomes, making adopting innovative strategies for sustainable waste management essential.

Predictive analytics leverages historical data and statistical algorithms to forecast future trends. In waste management, predictive analytics is a powerful tool for optimizing waste collection, improving resource allocation, and enhancing the overall efficiency of waste management systems. Studies have demonstrated that predictive analytics can lead to significant cost savings and improved service delivery. The benefits of predictive analytics include enhanced decision-making, as it provides actionable insights that enable waste management authorities to make informed decisions regarding collection schedules, resource allocation, and policy formulation (Aphale et al., 2015). Additionally, authorities can optimize collection routes and schedules by forecasting waste generation, reducing operational costs, and minimizing environmental impacts (Alsabt et al., 2024). Furthermore, predictive models can inform public awareness campaigns by identifying areas with the highest waste generation, allowing for targeted educational efforts (Amasuomo & Baird, 2016).

Research indicates that waste generation is closely linked to demographic, socio-economic, and cultural factors. As urban populations grow, so does the volume of waste produced. According to (Velis et al., 2023), cities in low-income countries produce significantly less waste per capita than high-income countries. Still, the rapid pace of urbanization in these regions is expected to exacerbate waste management challenges. Urbanization increases the quantity of waste generated and alters its composition. A shift toward more consumer-oriented lifestyles typically results in increased packaging and organic waste (Debrah et al., 2021). Studies show that urban areas with higher population densities tend to experience more significant waste generation per capita (Reis-Dennis, 2020).

Municipal solid waste (MSW) generation has evolved dramatically over the decades,

influenced by a confluence of factors, including demographic shifts, economic development, urbanization, technological advancements, and changing consumption patterns. Understanding these historical trends is crucial for developing effective waste management strategies in the face of growing urban populations and increasing waste volumes. Demographic factors are foundational in shaping waste generation patterns. As populations grow, particularly in urban areas, the volume of waste produced tends to increase. Historical data suggest that urbanization has been a significant driver of waste generation, as evidenced by the rapid growth of cities worldwide. The United Nations projects that by 2050, nearly 68% of the global population will reside in urban areas, up from 55% in 2018 (ESCAP, 2019) & (Graham, 2024). This urban migration is often accompanied by increased waste production due to heightened consumption behaviours associated with urban lifestyles.

The relationship between population density and waste generation is particularly noteworthy. Research indicates that cities with higher population densities generate more waste per capita compared to rural areas. A study by (Gritsch et al., 2024) found that densely populated urban areas in Europe produced significantly more waste, primarily due to increased packaging and consumer goods. As urban areas expand and become more populated, the infrastructure required for effective waste management often struggles to keep pace with the rising volumes of waste generated.

Economic development is another critical determinant of waste generation trends. Generally, waste generation tends to increase as countries transition from low-income to middle-income and eventually to high-income status. This pattern is evident in countries experiencing rapid economic growth, such as China and India. In developing countries, economic growth is often accompanied by a rise in consumption levels, leading to increased waste generation. A report by the (Morais et al., 2022) indicated that the waste generated in low-income countries is substantially lower than that in high-income countries, primarily due to lower consumption

rates. However, as these countries develop and urbanize, the volume of waste generated rises sharply. In contrast, high-income countries tend to produce more waste per capita, driven by higher consumption of goods and services.

The composition of waste also changes with economic development. As economies grow, the types of waste generated shift from organic materials and biodegradable waste to more complex waste streams, including plastics, electronics, and hazardous materials. This shift creates additional challenges for waste management systems, necessitating the development of more sophisticated recycling and disposal methods.

Urbanization profoundly impacts waste generation patterns, particularly in developing countries. Rapid urban growth leads to increased consumption of goods and services, resulting in higher waste production. As cities expand, the infrastructure needed to manage waste often lags, exacerbating waste management challenges. Cultural and socio-economic factors also influence the composition of waste generated in urban areas. The prevalence of single-use plastics and packaging materials in urban settings has increased significantly, driven by consumer convenience and marketing strategies. Research by (Vuppaladadiyam et al., 2024) demonstrated that urban areas are more likely to generate significant amounts of packaging waste, which poses challenges for recycling and waste management systems.

Furthermore, the shift toward a more consumer-oriented society has led to the generation of e-waste, a growing concern globally. Rapid technological advancements and the increasing reliance on electronic devices have resulted in a surge in e-waste generation, particularly in urban centres. A Global E-waste Monitor by (Forti et al., 2020) report indicated that global e-waste generation reached 53.6 million metric tons in 2019, with urban areas being the primary contributors.

Technological advancements have significantly influenced waste generation trends over the years. Introducing new materials, particularly plastics, has transformed the waste landscape. Since the mid-20th century, plastics have

become ubiquitous in packaging and consumer products, leading to increased waste generation. The convenience and versatility of plastics have made them popular, but their environmental impact is profound, contributing to significant amounts of non-biodegradable waste in landfills and the environment.

Advancements in production and packaging technologies have also contributed to waste generation. The shift toward online shopping and home delivery services has led to increased packaging waste, as products are often shipped with excessive packaging to ensure safety during transit. Research by the (Dokl et al., 2024) highlighted that global consumption of plastics is set to double by 2030, further exacerbating waste generation challenges.

Waste management practices have evolved in response to technological changes. Recycling technologies have advanced, enabling the recovery of materials from waste streams that were previously considered unrecyclable. However, the effectiveness of recycling programs often depends on public awareness and participation, highlighting the need for comprehensive education and outreach efforts.

Analysing historical trends in waste generation reveals several key patterns. In the early 20th century, waste generation was primarily composed of organic materials, such as food scraps and yard waste. As industrialization progressed and urban populations grew, the waste stream diversified, incorporating packaging materials, plastics, and other non-biodegradable materials.

In the 1960s and 1970s, concerns about pollution and waste management began to rise, leading to the establishment of regulatory frameworks to address these issues. The introduction of the Resource Conservation and Recovery Act (RCRA) in the United States in 1976 marked a significant turning point in waste management, emphasizing the importance of reducing waste generation and promoting recycling.

By the late 20th century, the recycling movement gained momentum, with many countries implementing policies to encourage recycling

and waste reduction. The introduction of curb side recycling programs and deposit-refund systems contributed to increased recycling rates and a shift in waste composition. Research indicates that recycling rates have steadily increased over the past few decades, although disparities remain between developed and developing countries.

Public policy and regulation play a critical role in shaping waste generation trends. Governments worldwide have implemented various strategies to address the challenges of increasing waste generation. Policies promoting recycling, reducing single-use plastics, and encouraging sustainable consumption are becoming increasingly common.

In recent years, many countries have adopted circular economy principles, which emphasize the importance of reducing waste generation and promoting resource efficiency. The European Union's Circular Economy Action Plan, adopted in 2020, aims to promote sustainable production and consumption patterns and reduce waste generation across member states. Additionally, international agreements, such as the Basel Convention, aim to regulate the movement of hazardous waste and promote environmentally sound waste management practices. These policies reflect a growing recognition of the need for coordinated efforts to tackle waste generation and its environmental impacts.

The implementation of extended producer responsibility (EPR) policies has gained traction in many countries, requiring manufacturers to take responsibility for the entire lifecycle of their products, including end-of-life disposal. This approach encourages businesses to design easier recycled products and reduces the burden on municipal waste management systems.

The integration of machine learning into waste management represents a transformative shift in how data is utilized to address waste-related challenges. Various machine learning techniques have been explored, each offering unique advantages for predicting waste generation and optimizing management strategies. Regression models are a cornerstone of predictive analytics in MSWM, providing a statistical framework for understanding the relationship between waste

generation and various influencing factors. These models can be broadly categorized into linear, multiple, and non-linear regression.

Linear regression establishes a linear relationship between a dependent variable (e.g., waste generation) and one or more independent variables. (Zhang et al., 2015) employed linear regression to analyse waste generation across various cities, discovering that population density and economic factors significantly influence waste volumes. (Dahlen & Lagerkvist, 2010) found that household size and income levels were significant predictors of waste generation rates in Sweden. Multiple regression extends linear regression by allowing researchers to simultaneously model the impact of several independent variables. (Zhang et al., 2019) applied multiple regression analysis to explore the relationships between socioeconomic factors and waste generation in urban China, revealing that income and educational attainment significantly affected waste generation patterns. Non-linear regression is employed when the relationship between the dependent and independent variables is not linear. (Majlessi et al., 2018) utilized non-linear regression techniques to model municipal solid waste generation in Tehran, achieving improved prediction accuracy by incorporating seasonal variations and economic indicators. Time series regression analyses historical data for trends, seasonality, and cyclical patterns. (Beigl et al., 2008) implemented time series regression to analyse waste generation data in Vienna, revealing significant seasonal trends that could inform waste management strategies.

Decision trees and random forests are popular machine-learning techniques widely used in analysing and predicting municipal solid waste generation. Decision trees model decisions and their possible consequences in a tree-like structure, allowing for straightforward interpretation. (Lin et al., 2021) utilized decision trees to analyse waste composition data in Chinese cities, revealing significant socio-economic influences on waste generation. (Solano Meza et al., 2019) applied decision trees to predict waste generation in urban Colombia, highlighting the impact of population density and public awareness. Random forests combine

multiple decision trees to improve prediction accuracy and robustness. (Zhang et al., 2022) employed random forests to enhance waste sorting efficiency, effectively predicting waste composition based on socio-economic factors. (Klafke et al., 2023) utilized random forests to analyse factors influencing waste generation in Brazil, achieving higher accuracy than traditional regression models.

Neural networks (NNs) have emerged as a powerful tool in MSWM, offering advanced capabilities for modelling complex relationships in waste generation data. (Noori et al., 2009) applied neural networks to predict municipal solid waste generation in Tehran, achieving high accuracy by capturing complex socio-economic relationships. (Kumar et al., 2017) utilized NNs to forecast waste generation in urban areas of India, effectively incorporating seasonal variations. (Fidelis et al., 2023) employed NNs to analyse municipal solid waste composition, enhancing sorting efforts through accurate classification based on socio-economic indicators. (Davis et al., 2023) explored NNs for optimizing waste collection routes in New York City, dynamically adjusting schedules based on real-time data from IoT sensors.

Various factors influence waste generation rates, and understanding these influences is crucial for developing effective management strategies. Demographic factors, such as population density, age distribution, and household composition, significantly shape waste generation patterns. Studies by (Jiboye et al., 2020) highlight how urban areas with higher population densities tend to generate more waste. Economic indicators, including economic development and income levels, also significantly impact waste generation. Higher-income levels are often associated with increased consumption and waste production (Zohoori & Ghani, 2017). Cultural attitudes toward waste disposal and recycling can influence waste generation, as research by (Gu et al., 2023) emphasizes the importance of public awareness campaigns in shaping waste management behaviours. Rapid urbanization increases the demand for goods and services, leading to higher waste production, particularly in developing countries. Additionally, the types of employment

and industries in a region can influence waste generation, as different sectors produce varying waste compositions.

Case studies from various developing countries provide valuable insights into the challenges and opportunities associated with MSWM. (Adefemi et al., 2023) highlighted inefficiencies in Lagos, Nigeria, and the potential for machine learning to improve operational efficiency. Similarly, (Islam et al., 2016) demonstrated the effectiveness of predictive modelling in optimizing waste collection strategies in Dhaka, Bangladesh. Technological advancements have revolutionized waste management practices globally. The integration of Geographic Information Systems (GIS) with machine learning has proven beneficial for optimizing waste collection routes. Studies by (Arefin et al., 2024) and (Vu et al., 2019) illustrate how GIS-based approaches can enhance spatial analysis. The Internet of Things (IoT) has emerged as a transformative technology in waste management. Research by (Hussain et al., 2024) showcases the potential of IoT-enabled environments through multiagent simulation, demonstrating their effectiveness in enhancing automation, real-time data processing, and decision-making in interconnected systems.

Effective public policy and regulatory frameworks are essential for promoting sustainable waste management practices. Research by (McDougall et al., 2008) emphasizes the importance of integrating waste management strategies into urban planning processes. Despite the promising potential of predictive analytics in waste management, several challenges hinder its widespread implementation, particularly in developing countries. Data quality and availability pose significant barriers, and the shortage of skilled personnel proficient in data analysis and machine learning techniques presents another challenge.

This study aims to evaluate the existing municipal solid waste management practices in Sheger City, Koye Sub-city, and leverage machine learning techniques to develop predictive models that accurately forecast waste generation trends, considering various socio-economic and demographic factors.

Materials and Methods

Study Area

Koye-Feche is a sub-city of Sheger City, situated approximately 25 kilometers east of Addis Ababa along the Adama Highway. Geographically, the sub-city is positioned between 08°30'00" E to 09°19'05" E latitude and 38°21'45" N to 39°10'54" N longitude. It occupies a strategic location within the larger metropolitan area of Addis Ababa, contributing to its urban dynamics. Koye-Feche shares boundaries with Addis Ababa to the west and northeast, Gelan Sub-city to the south, and the predominantly rural Adea district to the east.

According to the Sub-city Socio-Economic Profile (2022), Koye-Feche spans a total area of 18,725 hectares. The most recent master plan, developed in 2023, highlights that approximately 1,248.3 hectares of land within Koye-Feche have been designated for reserved purposes, while 6,241.6 hectares have already undergone urbanization. This substantial area of urbanized land is indicative of the rapid urban growth experienced by the sub-city in recent years.

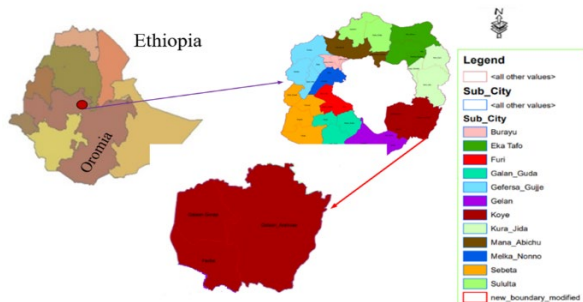


Figure 1. Location of Koye Feche Sub-City

The sub-city's geographical location and rapid urbanization have led to significant changes in its socio-economic structure. It has transformed from a predominantly rural area into a thriving urban center characterized by diverse residential, commercial, and industrial activities. These activities contribute to the generation of a wide range of MSW types. As the population grows and industrial development accelerates, waste

management challenges intensify, making Koye-Feche a critical region for studying waste generation patterns and exploring sustainable solutions.

Data Sources and Collection

The waste data were sourced from municipal waste management authorities within the Koye-Feche sub-city. They were collected through direct observations, waste audits, and reporting from waste management teams. Monthly and yearly waste data for each of the 14 waste types were recorded and compiled from 2009 to 2023.

Municipal staff conducted regular waste audits in designated areas of the sub-city, which provided the basis for estimating waste volumes. These audits involved measuring and categorizing the waste collected from residential, commercial, and industrial sectors. Additionally, official reports from the sub-city's waste management department provided supplementary data on waste generation trends, disposal methods, and recycling activities. The data were cross-verified for consistency to ensure reliability.

To enhance the accuracy of the data, a review of the records was conducted, where discrepancies, such as missing or inconsistent data, were addressed through consultations with local authorities and waste collection services. The collected data were then structured into a database with separate columns for each waste type, waste volume (tonnes), and period (monthly, weekly, and yearly) for each year in the dataset.

Waste Data Trend

Figure 2 illustrates the trends in total waste disposed, recycled, and generated, along with the recycling rate, in the study area from 2009 to 2023. The graph reveals a consistent upward trajectory in total waste generation (represented by the blue bars), which reflects the growing urbanization and increased consumption patterns in the region. The volume of waste generated has steadily risen over the years, driven by factors such as population growth and expanding industrial activities. Conversely, the amount of waste recycled (depicted by the green bars) has increased at a slower rate, with notable growth beginning around 2016. Despite this

increase in recycling, there remains a significant gap between the total waste generated and the waste recycled. The total waste disposed of (shown by the red bars) also follows an upward trend but at a rate slower than the total waste generated. This suggests that disposal strategies are expanding in parallel with waste generation but not at a rate sufficient to match recycling efforts or overall waste production. The black line representing the recycling rate (the ratio of recycled waste to total waste generated) demonstrates a positive trend, steadily rising from approximately 48% in 2009 to over 60% by 2023. This indicates an increasing efficiency in recycling efforts over time, reflecting improvements in waste management strategies, policy interventions, and public awareness. Notably, the recycling rate shows a sharp peak in 2019, which may suggest a significant policy or operational change aimed at boosting recycling. However, the overall growth rate of the recycling rate has somewhat slowed in the years following. These trends highlight the challenges posed by rising waste generation in the face of urban growth and the importance of expanding and optimizing recycling programs to keep pace with waste production.

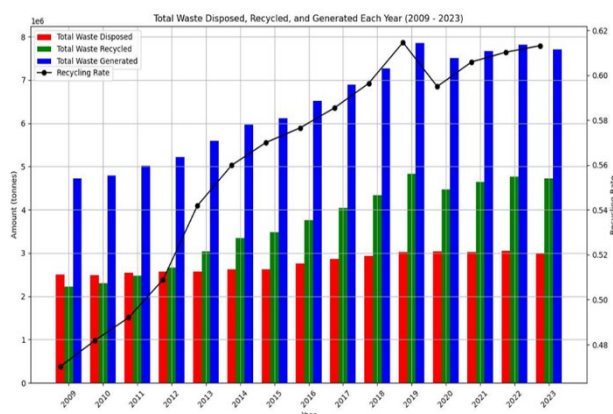


Figure 1. Illustrates the Trends in Total Waste Disposed, Recycled, and Generated

Methods

The methodology used to generate these waste generation forecasts combines three distinct machine learning approaches, which are described below

Autoregressive Integrated Moving Average

Autoregressive Integrated Moving Average (ARIMA) is a widely used statistical modelling technique for time series forecasting, rooted in the Box-Jenkins methodology. It combines three key components: autoregression (AR), differencing (I for integration), and moving average (MA) to model and predict future values based on historical data. The model is represented as ARIMA (p, d, q), where p denotes the order of the autoregressive terms (the influence of past values on the current value), d indicates the degree of differencing needed to achieve stationarity (removing trends or seasonality from the data), and q represents the order of the moving average terms (the impact of past forecast errors on the current value).

ARIMA's primary strength lies in its ability to capture linear patterns and trends in time series data, making it an excellent choice for forecasting tasks where the behaviour is consistent and predictable. This characteristic is beneficial in fields such as economics, finance, and environmental studies, including solid waste management. In the context of solid waste forecasting, ARIMA can be applied to predict the generation of municipal solid waste (MSW) or recycling rates over time based on historical waste data. By modelling waste generation patterns, ARIMA can help forecast future waste volumes, identify seasonal variations, and support decision-making processes related to waste collection, recycling, and disposal planning.

The flexibility of ARIMA allows it to be adapted for various types of waste data, including trends in specific waste streams such as food waste, plastics, and paper, as well as regional or temporal variations. For instance, the model can account for periodic fluctuations in waste generation tied to holidays, events, or climate patterns. However, while ARIMA excels in modelling linear trends, it is less effective at capturing nonlinear patterns or sudden, unforeseen shifts in waste production, such as those caused by policy changes or significant societal events. More advanced models like seasonal decomposition of time series or

machine learning approaches may be required in such cases.

Ultimately, ARIMA serves as a valuable tool for waste management professionals seeking to optimize waste handling operations and predict future trends. It contributes to more efficient resource allocation and improved sustainability practices in urban environments.

Random Forest

The research team of (Nisbet et al., 2009) and (Breiman, 1996), were pioneers in introducing the RF model-based bagging algorithm for classification tasks, utilizing random subspaces and bagging techniques. As one of the most influential machine learning algorithms, RF builds upon an ensemble of decision tree predictors. In this approach, each tree in the forest is constructed using a random subset of the data, where the feature values for each tree are sampled independently but according to the same distribution. This ensemble strategy allows the model to reduce overfitting by averaging the predictions from all individual trees, leading to a more robust and generalized output. The final prediction from the RF model is determined by aggregating the individual predictions of all trees, typically through averaging in regression tasks or voting in classification tasks (Shi & Horvath, 2006).

This ensemble approach provides a powerful mechanism to enhance model accuracy, improve stability, and mitigate the biases that may arise from a single decision tree, making RF one of the most popular and effective machine-learning techniques for both classification and regression problems. Furthermore, the model's ability to handle large datasets with high-dimensional features and its resilience against overfitting has contributed to its widespread use in various domains of machine learning.

Long Short-Term Memory

Long Short-Term Memory (LSTM) networks are specialized variants of Recurrent Neural Networks (RNNs) designed to address the vanishing gradient problem, a common challenge in traditional RNNs during backpropagation. Unlike standard RNNs, LSTMs introduce memory cells that enable the

network to retain and process information over extended periods. The core innovation in LSTMs lies in using three gating mechanisms: input, forget, and output gates that control the flow of information within the network. These gates determine the proportion of information retained in memory, discarded, or passed along to the next time step.

Compared to traditional RNNs, LSTMs are better equipped to handle long-term dependencies in sequential data. The gating mechanisms mitigate issues like vanishing gradients, which can hinder the training of conventional RNNs when working with long sequences. The input gate controls how much new information is added to the memory cell, the forget gate determines what information should be discarded, and the output gate decides which parts of the memory cell are used to generate the network's output at each time step. This selective retention and removal of information allows LSTMs to maintain relevant data over long sequences, making them highly effective for tasks requiring long-term memory.

LSTMs have proven particularly valuable in sequential data applications, such as time series forecasting, natural language processing, and speech recognition. Their ability to capture both short-term and long-term dependencies enables them to perform superior in many complex, real-world tasks. Additionally, LSTMs are widely used in domains like solid waste type, video analysis, machine translation, and stock market prediction, where understanding and remembering past information is crucial for making accurate predictions.

Model Testing and Validation

To assess the effectiveness of the regression algorithms for the intended objective, the trained models were evaluated using a separate dataset that was not involved in the training phase. The validation process was structured in two distinct stages: a parameter optimization phase followed by a training and prediction phase. During the parameter optimization stage, a grid search was performed in combination with tenfold cross-validation to identify the most suitable parameters for all tree models. This allowed for

a comprehensive tuning of the model's hyperparameters to maximize performance.

In the subsequent training and prediction phase, the optimized parameters were applied to the models, and the results were compared using various error metrics. The model's accuracy was assessed by comparing the training and test scores, providing an estimate of its generalizability. Additionally, the predictive performance of the machine learning models was quantified through key evaluation metrics, including Mean Squared Error (MSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2). These metrics provide a clear understanding of the model's error rates and its ability to explain the data variance, as (Handelman et al., 2019) detailed.

Table 1. Model validation results (averaged across all waste types)

Model	MSE	MAE	R^2
ARIMA	3.84×10^8	15,200	0.85
RF	2.91×10^8	12,800	0.89
LSTM	1.62×10^8	9500	0.93

$$MAE = \frac{\sum_{i=1}^n (Y_i - X_i)^2}{n} \quad (1)$$

Where Y_i is the predicted value, X_i is the actual value, and n is the total number of data points

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i)^2 \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{Y}_i - Y_i)^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \quad (3)$$

Where $\hat{Y}_i$ is the model predicted value of waste generation rate, Y_i is the actual (data) value and $\bar{Y}$ is the mean, and n is the number of observations.

The Sum of Squared Errors (SSE) was also calculated as a percentage value derived from the Root Mean Squared Error (RMSE). The SSE and R^2 were computed as key performance

indicators for both the training and testing datasets. Adjustments to the model parameters and structure typically resulted in lower training errors than testing errors.

Results and Discussion

The machine learning-driven forecasts for MSW generation from 2024 to 2028 illuminate a trajectory of sustained growth across the 14 identified waste categories, culminating in a projected total waste volume of 4,177,500 tonnes by 2028 according to the LSTM model, an 8.5% increase from the 3,852,150 tonnes recorded in 2023. This escalation is not uniformly distributed across waste types, with significant contributions from high-volume streams such as food waste, plastic waste, and paper/cardboard (P/Cd) collectively underscoring the mounting pressure on waste management systems. The LSTM model, leveraging its sequential learning capabilities, predicts a marked rise in food waste from 404,900 tonnes in 2023 to 420,800 tonnes in 2024, progressing steadily to 430,500 tonnes in 2025, 440,200 tonnes in 2026, 450,000 tonnes in 2027, and reaching 459,800 tonnes by 2028 a cumulative growth of 13.5%. This trajectory reflects an annualized increase of approximately 2.7%, potentially driven by intensifying urbanization, population expansion, and persistent inefficiencies in food supply chains, highlighting a critical challenge for sustainable waste management. In contrast, plastic waste, another dominant contributor, is forecasted to increase from 407,600 tonnes in 2023 to 418,900 tonnes in 2024, advancing to 425,200 tonnes in 2025, 431,500 tonnes in 2026, 437,800 tonnes in 2027, and peaking at 444,100 tonnes by 2028, an 8.9% rise over the period. This growth, averaging roughly 1.8% annually, signals the enduring prevalence of single-use plastics and packaging, compounded by historically low recycling rates (0.06 in 2023), which exacerbate disposal burdens. Conversely, more stable categories such as glass and scrap tyres exhibit modest increments: glass waste rises from 35,650 tonnes in 2023 to 37,000 tonnes in 2024, 37,500 tonnes in 2025, 38,000 tonnes in 2026, 38,500 tonnes in 2027, and 39,000 tonnes by 2028 (LSTM), a 5.2% increase, while scrap tyres

grow from 17,950 tonnes in 2023 to 19,100 tonnes in 2024, 19,700 tonnes in 2025, 20,300 tonnes in 2026, 20,900 tonnes in 2027, and 21,500 tonnes in 2028, a 5.9% rise. These slower growth rates suggest relatively consistent consumption patterns and effective waste-handling mechanisms for these materials, though their cumulative impact remains non-trivial.

A rigorous comparison of the three machine learning models ARIMA, RF, and LSTM over the 2024–2028 period reveals pronounced differences in their forecasting tendencies, with LSTM consistently projecting higher waste volumes, particularly for categories exhibiting nonlinear historical trends. For the aggregate total waste, LSTM estimates 3,952,100 tonnes in 2024, escalating to 4,008,600 tonnes in 2025, 4,064,900 tonnes in 2026, 4,121,200 tonnes in 2027, and 4,177,500 tonnes by 2028, reflecting an average annual increase of approximately 56,000 tonnes. RF, by comparison, predicts a more tempered rise from 3,916,900 tonnes in 2024 to 3,965,500 tonnes in 2025, 4,014,100 tonnes in 2026, 4,062,700 tonnes in 2027, and 4,111,300 tonnes in 2028, averaging 48,000 tonnes annually, while ARIMA offers the most conservative outlook, projecting 3,889,000 tonnes in 2024, 3,925,950 tonnes in 2025, 3,962,900 tonnes in 2026, 3,999,850 tonnes in 2027, and 4,036,800 tonnes in 2028, with an annual increment of about 37,000 tonnes. This divergence is particularly stark in dynamic waste streams like food and plastic. For food waste, LSTM's forecasts (420,800 tonnes in 2024 to 459,800 tonnes in 2028) consistently outpace RF's (415,200 tonnes in 2024 to 445,600 tonnes in 2028) and ARIMA's (410,500 tonnes in 2024 to 433,300 tonnes in 2028), with LSTM's projections suggesting a growth rate nearly double that of ARIMA by 2028 (13.5% vs. 7.0%). Similarly, plastic waste projections show LSTM advancing from 418,900 tonnes in 2024 to 444,100 tonnes in 2028, compared to RF's range of 413,500 tonnes to 431,100 tonnes and ARIMA's 410,200 tonnes to 420,600 tonnes, indicating LSTM's sensitivity to accelerating consumption patterns that RF's ensemble approach and ARIMA's linear framework struggle to capture fully. For stable categories

like cardboard (Cd), however, the models converge more closely: LSTM predicts a progression from 812,700 tonnes in 2024 to 837,500 tonnes in 2028 (4.1% growth from 2023's 804,650 tonnes), RF from 810,200 tonnes to 831,800 tonnes, and ARIMA from 808,900 tonnes to 825,900 tonnes, reflecting a shared ability to model the steady demand driven by e-commerce and packaging needs.

The validation of these models against historical data from 2021 to 2023 provides a robust foundation for interpreting their predictive efficacy, with the inclusion of MSE alongside MAE and the R^2 offering a comprehensive assessment of model accuracy. LSTM demonstrates superior performance across all metrics, achieving an MSE of 1.62×10^8 tonnes², an MAE of 9,500 tonnes, and an R^2 of 0.93, compared to RF's MSE of 2.91×10^8 tonnes², MAE of 12,800 tonnes, and R^2 of 0.89, and ARIMA's MSE of 3.84×10^8 tonnes², MAE of 15,200 tonnes, and R^2 of 0.85. The lower MSE for LSTM indicates its enhanced precision in minimizing large prediction errors, which is particularly critical for volatile categories such as hazardous waste (Hw) and plastic waste, where substantial deviations can have significant operational and environmental implications. For hazardous waste, LSTM forecasts a rise from 164,150 tonnes in 2023 to 170,500 tonnes in 2024, 174,000 tonnes in 2025, 177,500 tonnes in 2026, 181,000 tonnes in 2027, and 184,500 tonnes by 2028, a 12.4% increase while ARIMA projects a more restrained increase to 166,700 tonnes in 2024, 169,250 tonnes in 2025, 171,800 tonnes in 2026, 174,350 tonnes in 2027, and 176,900 tonnes in 2028 (7.8% growth), with its higher MSE reflecting greater error in capturing industrial or regulatory fluctuations. RF falls between these extremes, predicting 168,300 tonnes in 2024 to 179,900 tonnes in 2028, with its intermediate MSE suggesting moderate success in handling nonlinearity but less precision than LSTM. The R^2 values further confirm LSTM's explanatory power, with 93% of the variance in the test data explained, compared to 89% for RF and 85% for ARIMA, while the MAE highlights LSTM's ability to reduce average errors, reinforcing its suitability

for the diverse temporal patterns inherent in MSW generation.

These forecasts carry profound implications for waste management strategies over the 2024–2028 period, necessitating tailored interventions to address the escalating waste burden. Food waste, projected to reach 459,800 tonnes by 2028 (LSTM), poses a significant challenge, with its trajectory from 420,800 tonnes in 2024 to 450,000 tonnes in 2027, signalling a need for aggressive waste diversion measures. Composting programs, food recovery networks, and public education campaigns could mitigate the environmental impact, particularly given the category's modest recycling rate (0.16 in 2023), which translates to substantial landfill contributions (e.g., 353,472 tonnes disposed of in 2024 per LSTM, assuming constant rates). Plastic waste, rising from 418,900 tonnes in 2024 to 444,100 tonnes by 2028 (LSTM), presents an equally pressing issue, compounded by its persistently low recycling rate (0.06 in 2023), projecting disposal volumes from 393,766 tonnes in 2024 to 417,454 tonnes in 2028. This trend advocates for policy innovations such as EPR frameworks, bans on single-use plastics, or investments in chemical recycling technologies to curb landfill dependency. Aggregate waste projections further underscore the urgency, with LSTM's estimate of 4,177,500 tonnes by 2028 compared to 3,952,100 tonnes in 2024, necessitating expanded waste-to-energy facilities, enhanced recycling infrastructure, or regional landfill capacity planning to accommodate an additional 225,400 tonnes over the period. Stable categories like glass and scrap tyres, while growing modestly (e.g., glass from 37,000 tonnes in 2024 to 39,000 tonnes in 2028), contribute incrementally to the total waste stream, suggesting opportunities for optimizing existing recycling systems rather than overhauling them entirely.

Despite the robustness of these projections, several limitations warrant consideration. The models assume that historical patterns from 2009 to 2023 will persist through 2028, potentially underestimating the impact of exogenous shocks such as economic recessions, technological breakthroughs, or stringent waste reduction policies. For instance, a hypothetical

increase in recycling rates from 0.06 to 0.20 for plastic waste due to new regulations could reduce disposal volumes significantly, rendering current forecasts overly conservative. Conversely, unmodeled variables such as population growth, industrial output, or shifts in consumer behaviour could amplify waste generation beyond these estimates, particularly in categories like food and hazardous waste. The absence of these external covariates in the current models represents a methodological constraint, though their inclusion could enhance predictive precision in future iterations. Additionally, the reliance on a relatively short historical dataset (15 years) may limit the models' ability to detect long-term cyclical patterns, though LSTM's sequential architecture mitigates this to some extent by emphasizing recent trends. Future research could address these gaps by integrating socio-economic indicators (e.g., GDP, urbanization rates) or exploring hybrid models (e.g., ARIMA-LSTM) to combine the strengths of linear and nonlinear approaches, thereby refining the forecasts for both volatile and stable waste streams.

The MSW generation forecasts from 2024 to 2028 highlight a critical juncture for waste management planning, with LSTM emerging as the most reliable model for capturing the complex dynamics of waste production. Its projections from 3,952,100 tonnes in 2024 to 4,177,500 tons in 2028 offer a forward-looking perspective that surpasses RF's and ARIMA's more tempered estimates, particularly for high-growth categories like food (13.5% increase) and plastic (8.9% increase). This analysis quantifies the escalating waste challenge and underscores the need for proactive, category-specific strategies such as composting for food waste and advanced recycling for plastics while identifying opportunities for optimization in stable streams like glass and cardboard. By bridging predictive modelling with actionable insights, this study provides a robust framework for policymakers and practitioners to navigate the waste management landscape over the next five years, ensuring sustainability and resilience in the face of growing waste volumes.

Conclusion

This study emphasizes the growing challenges associated with MSWM in rapidly urbanizing areas. The findings underscore the importance of adopting advanced analytical techniques, such as machine learning, to address the complexities of waste generation, particularly in cities experiencing significant population growth and economic development. The three machine learning ARIMA, RF, and LSTM were used to predict waste generation trends in Sheger City, Koye Sub-city, Ethiopia, from 2024 to 2028. Among these models, LSTM demonstrated superior performance, achieving the lowest MSE, MAE, and R^2 , making it the most reliable tool for forecasting future waste generation. The model's accuracy highlights the ability of machine learning algorithms to capture complex relationships and temporal dependencies in waste data, providing valuable insights for waste management authorities.

The projections indicate that total waste generation in Sheger City will increase by 8.5%, rising from 3,852,150 tonnes in 2023 to 4,177,500 tonnes by 2028. High-growth waste streams, such as food waste (13.5% increase) and plastic waste (8.9% increase), represent substantial challenges for waste management systems. These increases are expected to put considerable pressure on current waste management infrastructure and highlight the need for more effective strategies in handling and recycling these high-volume waste categories.

The insights gained from this study call for targeted, data-driven interventions in waste management. For food waste, which is projected to grow significantly, interventions such as food recovery programs, composting initiatives, and public awareness campaigns can reduce environmental impact. Given the relatively low recycling rates for food waste, these efforts will be essential for diverting waste from landfills and minimizing its ecological footprint. In contrast, plastic waste, which exhibits significant growth, presents an even more urgent issue due to its low recycling rates and high environmental persistence. The study advocates for policies promoting extended producer responsibility,

bans on single-use plastics, and adopting innovative recycling technologies, such as chemical recycling. These measures are essential for mitigating the increasing pressure from plastic waste.

Additionally, the findings indicate that waste streams like glass and scrap tyres, while experiencing slower growth, still contribute incrementally to the overall waste volume. These stable waste types present opportunities for optimizing existing recycling systems rather than developing entirely new infrastructure. Efficient waste sorting, improved collection systems, and public participation can significantly enhance the recycling rates for these materials.

The study also underscores the importance of future research in enhancing the predictive power of waste management models. Although the LSTM model demonstrated high accuracy in this study, several limitations were identified. The model's reliance on historical data and the exclusion of some socio-economic variables such as GDP growth, technological advancements, and changes in consumer behaviour suggest that incorporating these factors into future models could improve the precision of predictions. Additionally, the relatively short period of the dataset (15 years) may limit the ability of the model to detect long-term cyclical trends. Future studies could address these gaps by integrating additional socio-economic indicators, considering exogenous shocks (e.g., economic crises or environmental regulations), and exploring hybrid models that combine the strengths of different machine learning techniques, such as ARIMA-LSTM models.

Another future work includes exploring more granular data collection methods, which could provide more detailed insights into waste generation trends at a local level. This would allow for more tailored and effective waste management strategies that address the unique needs of different areas within a city. Integrating real-time data from sensors and machine learning algorithms can also facilitate dynamic waste collection scheduling, further optimizing the operational efficiency of waste management systems. In conclusion, the results of this study

not only highlight the utility of machine learning models for predicting municipal solid waste generation but emphasize the critical need for adaptive and forward-thinking waste management policies. As urban populations continue to grow and consumption patterns evolve, cities need to adopt data-driven approaches to waste management. By leveraging the power of predictive modelling and machine learning, urban authorities can better plan for future waste management challenges, ensuring sustainability, resource efficiency, and reduced environmental impact. This research provides a foundation for policymakers, urban planners, and waste management professionals to develop smarter, more efficient waste management systems that can cope with the increasing demands of modern urban life.

Acknowledgement

The authors gratefully acknowledge the MOFCOM Scholarship Board and the Chinese government for their generous support, as well as the staff of the UNEP-Tongji Institute of Environment for Sustainable Development for their invaluable assistance and encouragement throughout this research.

Conflict of Interests

No conflict of interest.

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